Questions: Introduction to logic

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Summary

A selection of questions for the study guide introducing propositional logic.

Before attempting these questions, it is highly recommended that you read Guide: Introduction to logic.

Q1 Propositions

Boole's Burgers has recently opened. To help Mary and George with their advertising strategy, determine which of the following sentences are propositions.

1.1. "Boole's Burgers is a burger restaurant."

1.2. "Our burgers taste amazing!"

1.3. "All burgers in the restaurant cost more than £2."

1.4. "Please give me a cheeseburger."

1.5. "What are your burger toppings?"

1.6. "Some customers prefer vegan burgers."

1.7. "Are the fries sold out?"

1.8. "The fries are sold out."

Q2 Connectives

What is the logical structure for the following statements?

2.1. "If you get two burgers, you'll get a discount."

2.2. "I'll get one cheeseburger and one vegan burger."

2.3. "I won't get cheese on my burger."

2.4. "I'll get a hamburger, or I'll get a milkshake."

2.5. "I'll only get a cheeseburger if you get one too."

2.6. "If you go to Boole's Burgers, I'll go too, and I'll get a vegan burger."

Q3 Truth tables

Give the truth tables for the following logical statements:

3.1. $P \to Q$

3.2. $\neg P$

3.3. $(P \vee Q) \wedge \neg Q$

3.4. $\neg(P \leftrightarrow Q)$

3.5. $\neg(P \wedge Q)$

3.6. $(P \to Q) \to R$

3.7. $\neg(P \vee (Q \vee R))$

3.8. $(P \leftrightarrow Q) \vee (\neg R)$

Q4 Equivalence

Using truth tables or otherwise, are the following pairs of statements logically equivalent?

4.1. $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$

4.2. $P \to Q$ and $Q \to P$

4.3. $P \to Q$ and $(\neg P) \vee Q$

4.4. $P \to Q$ and $\neg(P \vee Q)$

After attempting the questions above, please click this link to find the answers.

Version history and licensing

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